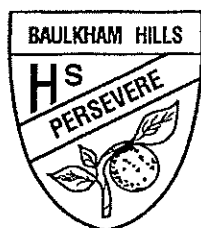


BAULKHAM HILLS HIGH SCHOOL

YEARLY EXAMINATION



YEAR 11

MATHEMATICS

ADVANCED

2010

General Instructions

- Exam time – 3 hours
- Reading time – 5 minutes
- Start each question on a new page
- All necessary working should be shown
- Write your student number at the top of each page of your answer booklet
- Board approved calculators may be used
- Write, using black or blue pen

Total Marks: 120

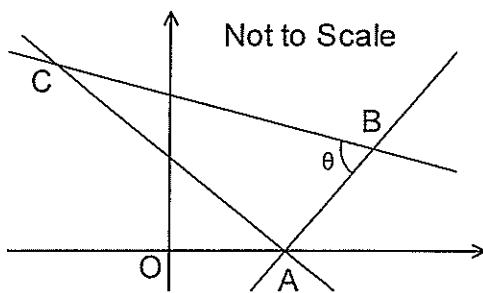
Attempt ALL questions

Question 1 (12 marks)

Marks

- | | | |
|----|---|---|
| a) | Evaluate, correct to 2 decimal places.
$19^{-0.5}$ | 2 |
| b) | Express $0.\dot{4}\dot{7}$ as a fraction in its simplest form. | 2 |
| c) | Solve $4 - 5x < 11$ | 2 |
| d) | Factorise $3x^2 - 5x - 2$ | 2 |
| e) | By showing full working, rationalise the denominator for the algebraic expression
$\frac{3}{\sqrt{x} + 2}$ | 2 |
| f) | Simplify $\frac{2}{3} - \frac{x-1}{4}$ | 2 |

Question 2 (12 marks) - Start a new page



Copy the diagram onto your writing pad.

The diagram shows the points $A(1,0)$, $B(4,1)$ and $C(-1,6)$ in the cartesian plane.
Angle ABC is θ

- | | | |
|----|--|---|
| a) | Show that A and C lie on the line $3x + y = 3$ | 2 |
| b) | Show that the gradient of AB is $\frac{1}{3}$ | 1 |
| c) | Show that the length of AB is $\sqrt{10}$ units | 1 |
| d) | Show that AB and AC are perpendicular | 1 |
| e) | Find $\tan \theta$ | 2 |
| f) | Find the equation of the circle with centre A that passes through B | 2 |
| g) | Point D lies on the interval AC such that $AD = AB$. Find the co-ordinates of D | 2 |
| h) | On your diagram, shade the region satisfying the inequality $3x + y \leq 3$ | 1 |

Question 3 (12 marks) - Start a new page

Marks

a) Differentiate the following

i) $y = (x^2 - 5)(2x + 1)$

2

ii) $y = (3x^3 - 7)^4$

2

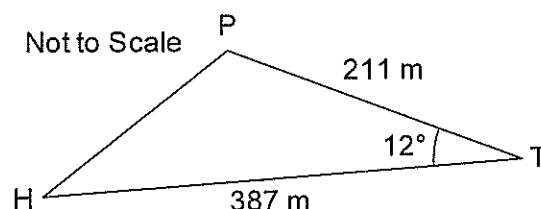
iii) $y = \frac{x^2 - 7}{2x + 3}$

2

b) Find the equation of the tangent to the curve $y = x^3 - 3x^2 + 3x - 1$ at the point where $x = 2$

3

c)

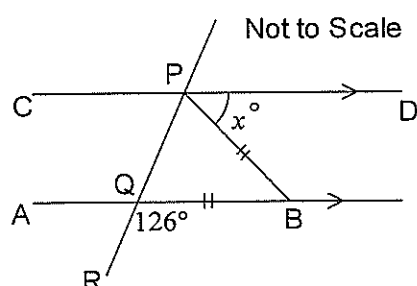


3

On a golf course, the distance from a tee T to the hole H is 387 metres.
A golfer's ball comes to rest at P, 211 metres from T.
Given that $\angle PTH = 12^\circ$ how far is it from P to H.

Question 4 (12 marks) - Start a new page

a)



In the diagram CD is parallel to AB
and $PB = QB$.
Copy the diagram onto your pad and
find the value of x° giving complete reasons.

3

b) The equation of a parabola is $x^2 = 12(y - 2)$

3

i) Find the co-ordinates of the vertex of the parabola.

ii) Find the equation of the directrix of the parabola.

c) For what value of k does $3x^2 + 2kx + 3 = 0$ have equal roots.

3

d) Find the equation of the normal to the curve $y = \sqrt{x + 2}$ at the point where $x = 7$

3

Question 5 (12 marks) - Start a new page

Marks

- a) Given $\sin \theta = \frac{3}{7}$ and that $\tan \theta < 0$ find the exact value of $\sec \theta$ for $0 \leq \theta \leq 360^\circ$ without finding θ and showing all working.

3

- b) If $x = \sqrt{a} - 1$ find $x^2 - \frac{1}{x}$

2

- c) State the domain and range for each function

i) $y = \sqrt{4 - x^2}$

2

ii) $y = \frac{1}{(x^2 - 1)}$

2

- d) Find the exact solution of

$$x = \frac{x+4}{x-1}$$

3

Question 6 (12 marks) - Start a new page

- a) Solve the simultaneous equations

$$3x - 2y = 6$$

$$4x + 5y = 31$$

3

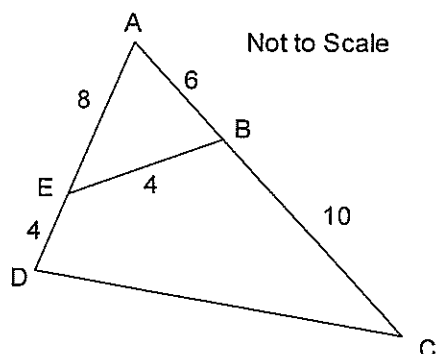
- b) Find the perpendicular distance from $P(7, -1)$ to the straight line $3x + 4y - 7 = 0$

3

- c) Express $\frac{6^n + 3^n}{2^{n+1} + 2}$ as a fraction in its simplest form.

2

- d)



In the diagram $AB = 6, BC = 10, AE = 8, ED = 4$

- i) Prove $\triangle ABE$ and $\triangle ADC$ are similar

2

- ii) Find the length of CD

2

Question 7 (12 marks) - Start a new page

Marks

- a) Without solving the equation $5x^2 + 3x + 6 = 0$

Find the value of

i) $\alpha + \beta$

1

ii) $\alpha\beta$

1

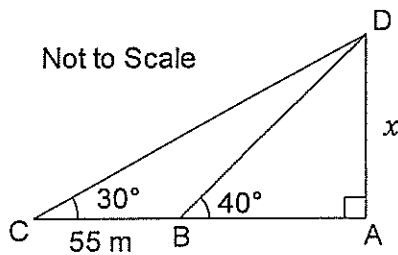
iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

2

iv) $\alpha^2 + \beta^2$

2

- b)



In the diagram $\angle DAC = 90^\circ$, $\angle DBA = 40^\circ$,
 $\angle DCA = 30^\circ$, $CB = 55 \text{ m}$ and $AD = x \text{ m}$

- i) Find DB

2

- ii) Hence find x

2

- c) Find the values of x for which $|3x - 2| \leq 6$

2

Question 8 (12 marks) - Start a new page

- a) Solve $4x^2 + 4x - 3 = 0$ by the method of completing the square.

3

- b) Prove that $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \sec \theta \operatorname{cosec} \theta$

3

- c) Solve the following equation

$$9^x - 12(3^x) + 27 = 0$$

3

- d) The point $P(x, y)$ moves so that its distance AP from the point $A(5, 1)$ is always twice its distance BP from the point $B(-1, 4)$.

3

Show that the equation of the locus is

$$x^2 + y^2 + 6x - 10y + 14 = 0$$

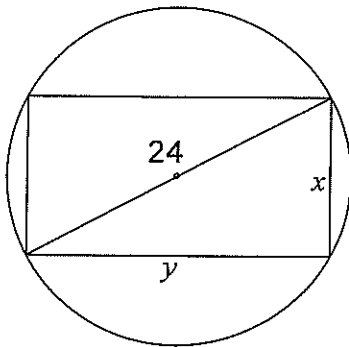
Question 9 (12 marks) - Start a new page

Marks

- a) Consider the function $f(x) = 3x^4 - 4x^3$
- Find the co-ordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 4
 - Find the co-ordinates of any points of inflexion 2
 - Find the greatest and the least values of $y = f(x)$ in the domain $-1 \leq x \leq 2$ 2
 - Draw a neat sketch of the curve $y = f(x)$ for $-1 \leq x \leq 2$ 2
- b) Express $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta$ as a single trigonometric ratio 2
- and hence solve $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = 1$ for $0 \leq \theta \leq 360^\circ$

Question 10 (12 marks) - Start a new page

a)



A rectangle of sides x cm and y cm is contained in a circle of diameter 24cm as shown

- Show that the area A of the rectangle is given by $A = x\sqrt{576 - x^2}$ 2
 - Find $\frac{dA}{dx}$ 2
 - Find the area of the largest rectangle that can be drawn in this circle. 3
- b) Prove that $\frac{1}{\sec x + \tan x} = \sec x - \tan x$ 3
- c) Make y the subject if $2\sqrt{x}\sqrt{y} - y + x^2 = x$ 2

End of Paper

YR 11 ADVANCED SOLUTIONS YEARLY 2010

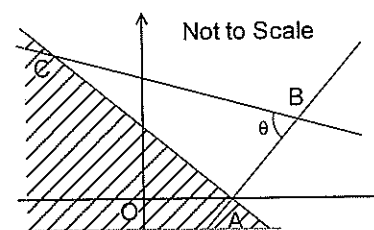
Question 1 (12 marks)

a)	$= 0.229415...$ $= 0.23$	2
b)	$n = 0.4747...$ $100n = 47.4747...$ $99n = 47$ $n = \frac{47}{99}$	2
c)	$-5x < 7$ $x > -\frac{7}{5}$	2
d)	$(3x+1)(x-2)$	2
e)	$\frac{3}{\sqrt{x}+2} \times \frac{\sqrt{x}-2}{\sqrt{x}-2} = \frac{3\sqrt{x}-6}{x-4}$	2
f)	$\frac{2 \times 4 - 3(x-1)}{12} = \frac{8-3x+3}{12}$ $= \frac{11-3x}{12}$	2

Question 2 (12 marks)

a)	$3x + y = 3$ subst $x = 1, y = 0$ $LHS = 3 + 0 = 3 = RHS$ subst $x = -1, y = 6$ $\therefore 3 \times -1 + 6 = -3 + 6$ $= 3 = RHS$ $\therefore A \text{ and } C \text{ lie in } 3x + y = 3$	2
b)	$m_{AB} = \frac{1-0}{4-1} = \frac{1}{3}$	1
c)	$AB = \sqrt{(4-1)^2 + (1-0)^2}$ $= \sqrt{10}$	1
d)	$m_{AC} = \frac{6-0}{-1-1} = -3$ $m_{AC} \times m_{AB} = -3 \times \frac{1}{3} = -1$ $\therefore AB \text{ and } AC \text{ are perpendicular}$	1
e)	$\tan \theta = \frac{AC}{AB} = \frac{\sqrt{2^2 + 6^2}}{\sqrt{10}} = 2$	2
f)	$(x-1)^2 + y^2 = (\sqrt{10})^2$ $x^2 + y^2 - 2x - 9 = 0$	2
g)	$\tan \theta = 2 \quad AC = 2AB$ $\therefore D \text{ is midpoint of } AC$ $\therefore D(0,3)$	2

h)



1

Question 3 (12 marks)

a)	i) $y' = 2x(2x+1) + x^2 - 5(2)$ $= 6x^2 + 2x - 10$ ii) $y' = 4(3x^3 - 7)^3 \times 9x^2$ $= 36x^2(3x^3 - 7)^3$ iii) $y' = \frac{2x(2x+3) - (x^2-7) \times 2}{(2x+3)^2}$ $= \frac{4x^2 + 6x - 2x^2 + 14}{(2x+3)^2}$ $= \frac{2x^2 + 6x + 14}{(2x+3)^2}$	2
b)	At $x = 2, y = 8 - 12 + 6 - 1$ $= 1$ $y' = 3x^2 - 6x + 3$ at $x = 2, y' = 12 - 12 + 3 = 3$ $\therefore \text{tangent is}$ $y - 1 = 3(x - 2)$ $y = 3x - 5$	3
c)	$PH^2 = PT^2 + HT^2 - 2 \cdot PT \cdot HT \cdot \cos \angle PTH$ OR $(q^2 = p^2 + h^2 - 2ph \cos T)$ $PH^2 = 211^2 + 387^2 - 2 \times 211 \times 387 \cos 12^\circ$ $PH^2 = 34544.80273$ $PH = 185.862322...$	3

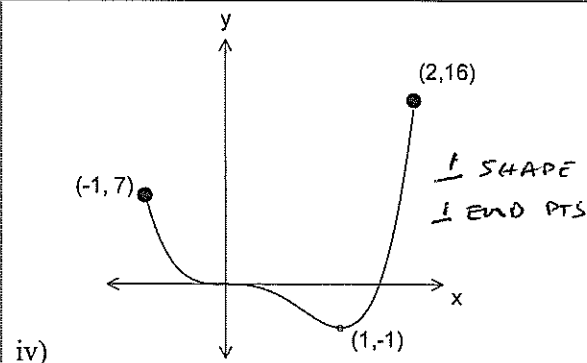
Question 4 (12 marks)

a)	 $\angle PQB + \angle BQR = 180^\circ$ ($\angle$ sum of a line) $\therefore \angle PQB = 180 - 126 = 54^\circ$ $\angle BPQ = \angle PQB = 54^\circ$ (base $\angle$ of isos Δ) $\angle DPQ = \angle BQR = 126^\circ$ (corresp $\angle$ on $//$ lines) $\therefore \angle DPB = \angle DPQ - \angle BPQ$ $= 126 - 54 = 72^\circ$	3
b)	$x^2 = 4 \times 3(y - 2)$ i) $V(0,2)$ ii) equation of directrix is $y = 2 - 3 = -1$ $y = -1$	3

c)	$\Delta = 4k^2 - 36 = 0$ $k = \pm 3$	3	(2 matching sides in the same ratio and included $\angle$ equal)	
d)	$x = 7, y = 3$ $m_T = \frac{1}{2}(x+2)^{-\frac{1}{2}}$ $x = 7 \quad m_T = \frac{1}{2\sqrt{9}} = \frac{1}{6}$ $\therefore m_n = -6$ $y - 3 = -6(x - 7), \quad 6x + y - 45 = 0$	3	ii) $\frac{EB}{DC} = \frac{1}{2}$ (matching sides in similar Δ 's are =) $\frac{4}{DC} = \frac{1}{2}$ $\therefore DC = 8$	2
Question 5 (12 marks)				
a)	$\sin \theta > 0 \quad \tan \theta < 0$ $\therefore$ 2nd quad $\cos \theta = -\frac{\sqrt{40}}{7}$ $\therefore \sin \theta = -\frac{7}{\sqrt{40}}$	3		
b)	$x^2 - \frac{1}{x} = (\sqrt{a} - 1)^2 - \frac{1}{\sqrt{a} - 1} \times \frac{\sqrt{a} + 1}{\sqrt{a} + 1}$ $= a - 2\sqrt{a} + 1 - \left(\frac{\sqrt{a} + 1}{a - 1}\right)$	2		
c)	i) $D: x \leq 2$ or $-2 \leq x \leq 2$ $R: y \geq 0$	2		
	ii) $D: \text{all } x \text{ except } x = \mp 1$ $R: y > 0, y < -1$	2		
d)	$x(x-1) = x+4$ $x^2 - 2x - 4 = 0$ $x = \frac{x^2 \pm \sqrt{4+16}}{2}$ $x = 1 \pm \sqrt{5}$	3		
Question 6 (12 marks)				
a)	$(1) \times 5 \rightarrow 15x - 10y = 30 \rightarrow (3)$ $(2) \times 2 \rightarrow 8x + 10y = 12 \rightarrow (4)$ $(3) + (4) \rightarrow 23x = 92$ $x = 4, y = 3$	3		
b)	$d = \left \frac{7 \times 3 + -1 \times 4 - 7}{\sqrt{3^2 + 4^2}} \right $ $d = 2$	3		
c)	$\frac{3^n \cdot 2^n + 3^n}{2(2^n + 1)} = \frac{3^n(2^n + 1)}{2(2^n + 1)}$ $= \frac{3^n}{2}$	2		
d)	i) $\frac{AE}{AC} = \frac{8}{16} = \frac{1}{2}$ $\frac{AB}{AD} = \frac{6}{12} = \frac{1}{2}$ $\therefore AE:AC = AB:AD$ $\angle DAC$ is common $\therefore \triangle ABE \sim \triangle ADC$	2		
Question 7 (12 marks)				
a)	i) $\alpha + \beta = -\frac{3}{5}$ ii) $\alpha\beta = \frac{6}{5}$ iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = -\frac{3}{6} = -\frac{1}{2}$ iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= \left(-\frac{3}{5}\right)^2 - 2 \times \frac{6}{5}$ $= -\frac{51}{25}$	1 1 2 2		
b)	i) $\angle CDB = 10^\circ$ (ext $\angle =$ sum interior opp $\angle$'s) $\frac{DB}{\sin 30^\circ} = \frac{CB}{\sin 10^\circ}$ $DB = 158.366188...$ ii) $\sin 40^\circ = \frac{x}{DB}$ $x = 101.795823...$	2 2		
c)	$-6 \leq (3x - 2) \leq 6$ $-4 \leq 3x \leq 8$ $-\frac{4}{3} \leq x \leq \frac{8}{3}$	2		
Question 8 (12 marks)				
a)	$x^2 + x - \frac{3}{4} = 0$ $\left(x^2 + x - \frac{1}{4}\right) - \frac{3}{4} = \frac{1}{4}$ $\left(x + \frac{1}{2}\right)^2 = 1$ $x = -\frac{1}{2} \pm 1$ $x = -\frac{3}{2}$ or $\frac{1}{2}$	3		
b)	$LHS = \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$ $= \frac{1}{\cos \theta \sin \theta}$ $= \sec \theta \csc \theta$ $= RHS$	3		
c)	$y = 3^x$ $\therefore y^2 - 12y + 27 = 0$ $y = 9$ or $y = 3$ $x = 1$ or 2	3		

~~$$\therefore 3^x = 9 \text{ or } 3^x = 1$$
$$x = 1 \text{ or } 2$$~~

d) $AP = 2PB$
 $AP^2 = 4PB^2$ 1
 $(x-5)^2 + (y-1)^2 = 4(x+1)^2 + 4(y-4)^2$
 $x^2 - 10x + 25 + y^2 - 2y + 1 = 4x^2 + 8x + 4 + 4y^2 - 32y + 64$ 1
 $\therefore 3x^2 + 3y^2 - 18x - 30y + 42 = 0$
 $\therefore x^2 + y^2 - 6x - 10y + 14 = 0$ 1



b)	$\frac{\sin^3 \theta}{\cos \theta} + \frac{\sin \theta \cos^2 \theta}{\cos \theta}$ $= \frac{\sin^2 \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$ $= \tan \theta$ $\tan \theta = 1$ $\theta = 45^\circ \text{ or } 225^\circ$	2
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Question 9 (12 marks)

a)	i) $f(x) = 3x^4 - 4x^3$ $f'(x) = 12x^3 - 12x^2$ $f''(x) = 36x^2 - 24x$ $f'(x) = 0 \therefore 12x^3 - 12x^2 = 0$ $12x^2(x - 1) = 0$ $x = 0 \text{ or } 1$
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x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$f'(x)$			

must show values

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$f'(x)$			

$\therefore (0,0)$ is a horizontal inflexion
 $(1,-1)$ is a min

$$\begin{aligned} \text{ii) } f''(x) &= 0 \\ 36x^2 - 24x &= 0 \\ 12x(3x - 2) &= 0 \quad \underline{1} \\ x = 0, \quad y &= 0 \text{ horizontal inflexion} \\ x = \frac{2}{3} \quad f''\left(\frac{2}{3}\right) &= 0 \\ f''(1) &> 0 \\ f''\left(\frac{1}{3}\right) &< 0 \\ \therefore x = \frac{2}{3} \quad y &= 3\left(\frac{2}{3}\right)^4 - 4\left(\frac{2}{3}\right)^3 \\ &= -\frac{16}{27} \end{aligned}$$

 $\left(\frac{2}{3}, -\frac{16}{27}\right)$ inflexion

iii) $f(-1) = 7$ $f(2) = 16$
greatest is 16
least is -1

Question 10 (12 marks)

a) area of rectangle = xy

But $x^2 + y^2 = 24^2$

$$\therefore y = \sqrt{576 - x^2} \quad \text{1}$$
$$\therefore A = x\sqrt{576 - x^2}$$
$$= x(576 - x^2)^{\frac{1}{2}} \quad \text{1}$$
$$\frac{dA}{dx} = (576 - x^2)^{\frac{1}{2}} + x \cdot \frac{1}{2} \cdot -\frac{2x}{(576 - x^2)^{\frac{1}{2}}} \quad \text{3}$$
$$= \frac{(576 - x^2) - x^2}{\sqrt{576 - x^2}}$$
$$= \frac{576 - 2x^2}{\sqrt{576 - x^2}} \quad \text{1}$$
$$= 0 \quad \text{min MAX} \quad \text{1}$$
$$\therefore x = 12\sqrt{2} \quad y = 12\sqrt{2} \quad \text{1}$$
$$\therefore A = 288 \text{ sq metres} \quad \text{1}$$

b)
$$\begin{aligned} LHS &= \frac{1}{\sec x \tan x} \times \frac{\sec x - \tan x}{\sec x - \tan x} \\ &= \frac{\sec^2 x - \tan^2 x}{\sec x - \tan x} \\ &= \frac{1}{\sec x - \tan x} \\ &= \sec x - \tan x = RHS \end{aligned}$$

c)	$2\sqrt{x}\sqrt{y} - y + x^2 = x$ $y - 2\sqrt{x}\sqrt{y} + x = x^2$ $(\sqrt{y} - \sqrt{x})^2 = x^2$ $\sqrt{y} - \sqrt{x} = \pm x$ $\sqrt{y} = \sqrt{x} \pm x$ $y = (\sqrt{x} \pm x)^2 \quad \underline{\underline{2}}$	2
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End of Exam